

Logistics Analytics Database — Snowflake Schema (ERD Description)

Schema Type

Snowflake schema: one fact table surrounded by dimension tables, several of which are further normalized into second-level "outrigger" dimensions (**Dim_Locations** is referenced by three different dimensions, and **Dim_Routes** references it twice for origin/destination — this is what distinguishes a snowflake from a plain star schema).

Grain

Fact_Shipments: one row = one shipment.

Entity-Relationship Diagram (text form)



Dim_Warehouses --(WarehouseID)- --(RouteID)-- FACT_SHIPMENTS

| \ / |

(WarehouseID) \ / (CarrierID)

| \ / |

Dim_Employees --(EmployeeID)-----\ /---(VehicleID)-- Dim_Vehicles

|

(CarrierID)

|

Dim_Carriers

Table Relationships (1-to-many unless noted)

Parent (dimension)	Child	Relationship
Dim_Locations	Dim_Customers	1 location -> many customers
Dim_Locations	Dim_Warehouses	1 location -> many warehouses
Dim_Locations	Dim_Routes (origin)	1 location -> many routes as origin
Dim_Locations	Dim_Routes (destination)	1 location -> many routes as destination
Dim_Carriers	Dim_Vehicles	1 carrier -> many vehicles
Dim_Warehouses	Dim_Employees	1 warehouse -> many employees
Dim_Dates	Fact_Shipments	1 date -> many shipments
Dim_Customers	Fact_Shipments	1 customer -> many shipments

Parent (dimension)	Child	Relationship
Dim_Products	Fact_Shipments	1 product -> many shipments (per line)
Dim_Carriers	Fact_Shipments	1 carrier -> many shipments
Dim_Vehicles	Fact_Shipments	1 vehicle -> many shipments
Dim_Warehouses	Fact_Shipments	1 warehouse -> many shipments
Dim_Employees	Fact_Shipments	1 employee -> many shipments (nullable)
Dim_Routes	Fact_Shipments	1 route -> many shipments

Why Snowflake (not Star)?

- **Dim_Locations** is a shared outrigger dimension reused by **Dim_Customers**, **Dim_Warehouses**, and twice by **Dim_Routes** — avoids repeating City/State/Region/Country text on every dimension row.
- **Dim_Vehicles** normalizes out from **Dim_Carriers** rather than embedding carrier attributes on every vehicle row.
- **Dim_Employees** normalizes out from **Dim_Warehouses** rather than embedding warehouse attributes on every employee row.

Row Counts (approximate, as generated)

Table	Rows	Columns
Fact_Shipments	~22,110	30
Dim_Dates	1,096	10
Dim_Customers	3,000	6
Dim_Products	800	7
Dim_Employees	900	5
Dim_Vehicles	500	6
Dim_Routes	350	6

Table	Rows	Columns
Dim_Warehouses	30	6
Dim_Locations	25	6
Dim_Carriers	10	5
Total	~29,821	87 documented columns

Total exceeds the brief's minimums (10 tables, 50+ columns, 20,000+ rows).